

A short note on cubic spline interpolation

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1 Introduction

We start by formally defining a spline function of degree $k - 1 \geq 0$.

Definition 1.1. A spline function $S(x)$ of degree $k - 1 \geq 0$ on a grid:

$$\Delta = \{a = x_0 < x_1 < \dots < x_n = b\}$$

of distinct knots is a real function with the following properties:

1. For x in $[x_i, x_{i+1}]$, $S(x)$ is a polynomial of degree $< k$.
2. Its first $k - 2$ derivatives are continuous, that is $S(x) \in C^{k-2}[a, b]$.

The space of all spline functions of degree $k - 1 \geq 0$ is denoted by $S_{\Delta, k}$. From the definition, it follows that if $S_1(x)$ and $S_2(x)$ are spline functions, so is $c_1 S_1(x) + c_2 S_2(x)$. Clearly, $S_{\Delta, k}$ is a linear vector space.

2 Cubic Spline Interpolant

A cubic spline uses cubic polynomials of degree 3 to interpolate between successive knots. A cubic spline interpolant $S(x)$ satisfies the following conditions:

1. Each piece or segment $S_j(x)$ on the interval $[x_j, x_{j+1}]$ is a cubic polynomial.
2. $S_j(x_j) = y_j$ for each $j = 0, 1, \dots, n$.
3. $S_j(x_{j+1}) = S_{j+1}(x_{j+1})$, $S'_j(x_{j+1}) = S'_{j+1}(x_{j+1})$ and $S''_j(x_{j+1}) = S''_{j+1}(x_{j+1})$.

Condition (3) is just a translation of the requirement that the whole spline should be $k - 2$ times differentiable and these derivatives must be continuous. As a crude approximation, we know that a continuous curve is one that can be drawn without lifting pen from paper. The individual segments of $S(x)$ are cubics, so they are continuous themselves. The only consideration is the continuity at knot points.

For a curve to be continuous at any given knot-point c , whether we approach it from the left or we approach it from the right, the spline function $S(x)$ must approach $S(c)$ regardless. Suppose the left piece is S_j and the right piece is S_{j+1} . This motivates $S_j(x_{j+1}) = S_{j+1}(x_{j+1})$. A similar argument follows for the first and second derivatives because we would like those to be continuous.

3 Construction of the Cubic Spline

To construct the cubic spline interpolant on a grid Δ divided into n subintervals, we require n segments $S_0, S_1, \dots, S_{n-1}$. In line, with condition (1), the segment S_j is a cubic polynomial of the form:

$$S_j(x) = a_j + b_j(x - x_j) + c_j(x - x_j)^2 + d_j(x - x_j)^3$$

and has 4 constants. Therefore, we have to determine the values of $4n$ constants.